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Abstract:

Brain tumor is the most common kind of cancer, accounting for millions of fatalities worldwide. Early detection and treatment of new dangerous Brain tumor cases are critical to ensuring a low mortality rate as well as a high survival rate. The majority of relevant research has focused on machine learning-based algorithms, but they haven't been able to achieve the highest levels of accuracy and specificity. To address this issue, this work employs a deep learning based convolutional neural network (CNN) classification mechanism based on advanced deep learning. For effective identification of the location of Brain tumor, a k-means clustering based segmentation technique is first applied. Finally, CNN was constructed for Brain tumor classification with grey level co-occurrence matrix (GLCM) based Texture features, decimated wavelet Transform (DWT) based low level features, and Statistical Color features, respectively, to archive the system's maximum efficiency. As a result, the research findings can be utilized to classify benign and malignant based brain tumors. In comparison to state-of-the-art techniques, simulation study demonstrates that the suggested method provides better qualitative and quantitative analyses.

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